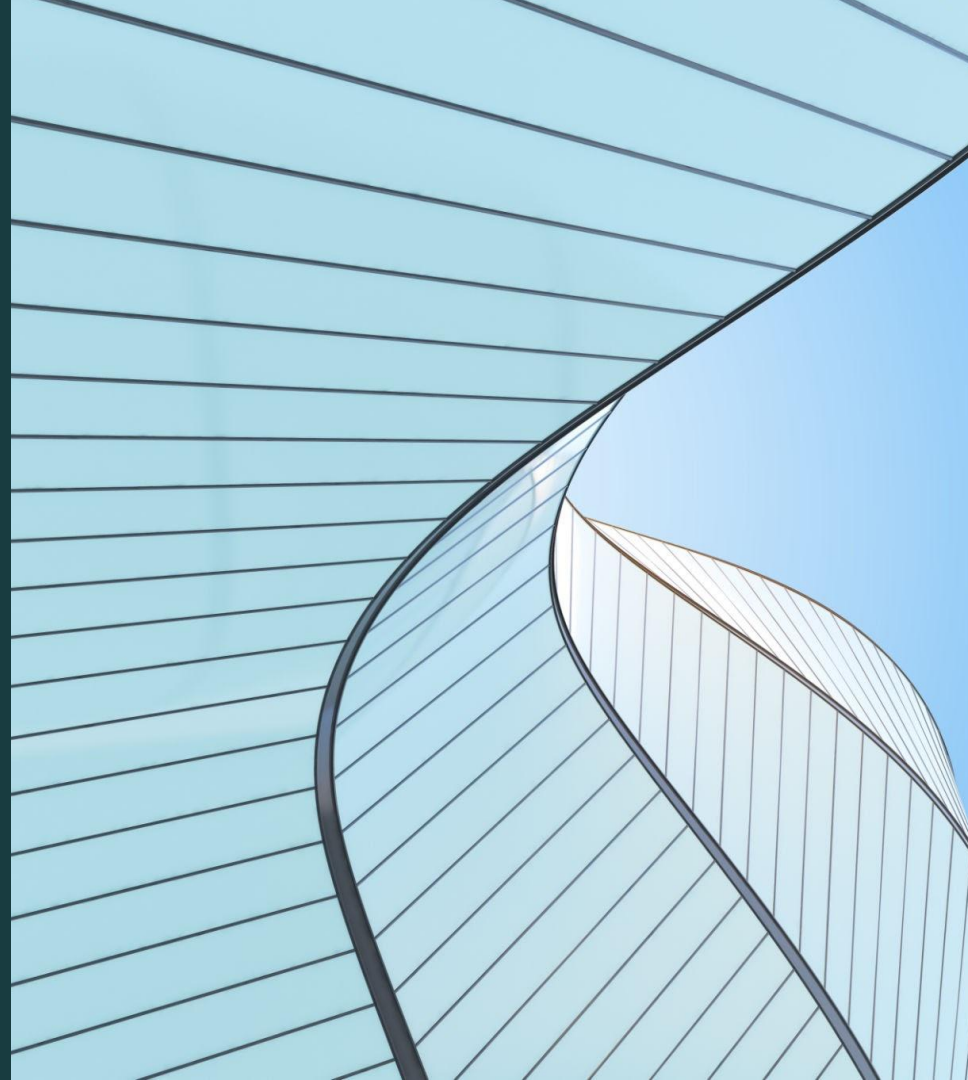


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AI- Powered Student Dropout *Prediction System*

Early Warning System for Improving Student Retention

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CONTEXT

Student dropout is not a new problem, but what's critical is that institutions often detect it *too late*.

There are signals early on—academic performance, financial issues—but they are *not systematically used*. This creates an opportunity for a *predictive, data-driven approach*.

THE BUSINESS PROBLEM

High Dropout Rates

Institutions face persistently high student attrition rates that impact their bottom line and mission

Revenue Loss

Lost tuition revenue from students who leave before completing their degree programs

Limited Early Warning

Current systems struggle to identify at-risk students early enough to intervene effectively

Lower Graduation

Reduced graduation rates that affect institutional reputation and accreditation metrics



THE APPROACH

We use data to identify at-risk students early and enable timely intervention

01

Analyze Historical Data

Examine past student records including grades, attendance, and financial information

02

Train Machine Learning Model

Build predictive algorithms that learn patterns and historical outcomes

03

Predict Risk Early

Identify students at risk of dropping out before it's too late to intervene

04

Enable Proactive Support

Connect at-risk students with resources and interventions that improve retention

KEY FINDINGS

What drives students dropout?
Our analysis revealed the most influential factors:

Academic Performance

The strongest predictor of dropout risk. Students with lower grades and course approval rates face significantly higher attrition

Financial Instability

Unpaid tuition and debtor status significantly increase dropout probability, creating immediate barriers to continued enrollment

Student Age

Older students show moderate influence in risk patterns, often balancing education with work and family responsibilities

MODEL PERFORMANCE

These metrics demonstrate the *model can detect* most at-risk students early, enabling timely interventions that improve *retention* outcomes.

90%

Model Accuracy

81%

Students at-risks
identified

93%

Predictive Reliability

The model can reliably separate students who are likely to drop out from those who are not.

DRIVERS OF DROPOUT RISK

Understanding the key factors that signal *high-risk* students

Low Course Approval Rates

Students failing to pass course face dramatically increased dropout probability and loss of momentum

Unpaid Tuition Fees

Outstanding balances crease immediate barriers to continued enrollment and academic progress

Debtor Status

Financial obligations to the institutions correlate strongly with increased attrition risk

RECOMMENDATIONS

Actionable *insights* in improving student retention

Monitor Academic Performance

Track grades and course approvals from the first semester to identify struggling students early

Financial Assistance Programs

Provide emergency aid, payment plans, and scholarships to reduce financial barriers

Build Early Alert Dashboards

Create real-time monitoring systems that flag at-risk students for immediate follow-up



RISKS AND CONSIDERATIONS

Important factors to keep in mind when *implementing* this system

Historical Data Dependency

Model is trained on past student behavior and may not perfectly predict future patterns

Institutional Generalization

Results may vary across different types of institutions with unique student populations

Potential Bias

Financial indicators could disproportionately affect low-income students if not carefully managed

Continuous Monitoring

Requires regular model updates and validation to maintain accuracy over time

NEXT STEPS AND RETURN ON INVESTMENTS

01

Deploy Model

Integrate into existing LMS and student information systems

Increased Retention

Proactive interventions prevent dropouts and keep students enrolled

02

Student Dashboards

Connect predictions with advisor and student-facing interfaces

Improved Graduation

Higher completion rates strengthen institutional reputation and outcomes

03

Pilot Program

Test intervention strategies with identified at-risk cohorts

Better Support

Data-driven insights enable more effective student success initiatives



Thank you!

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